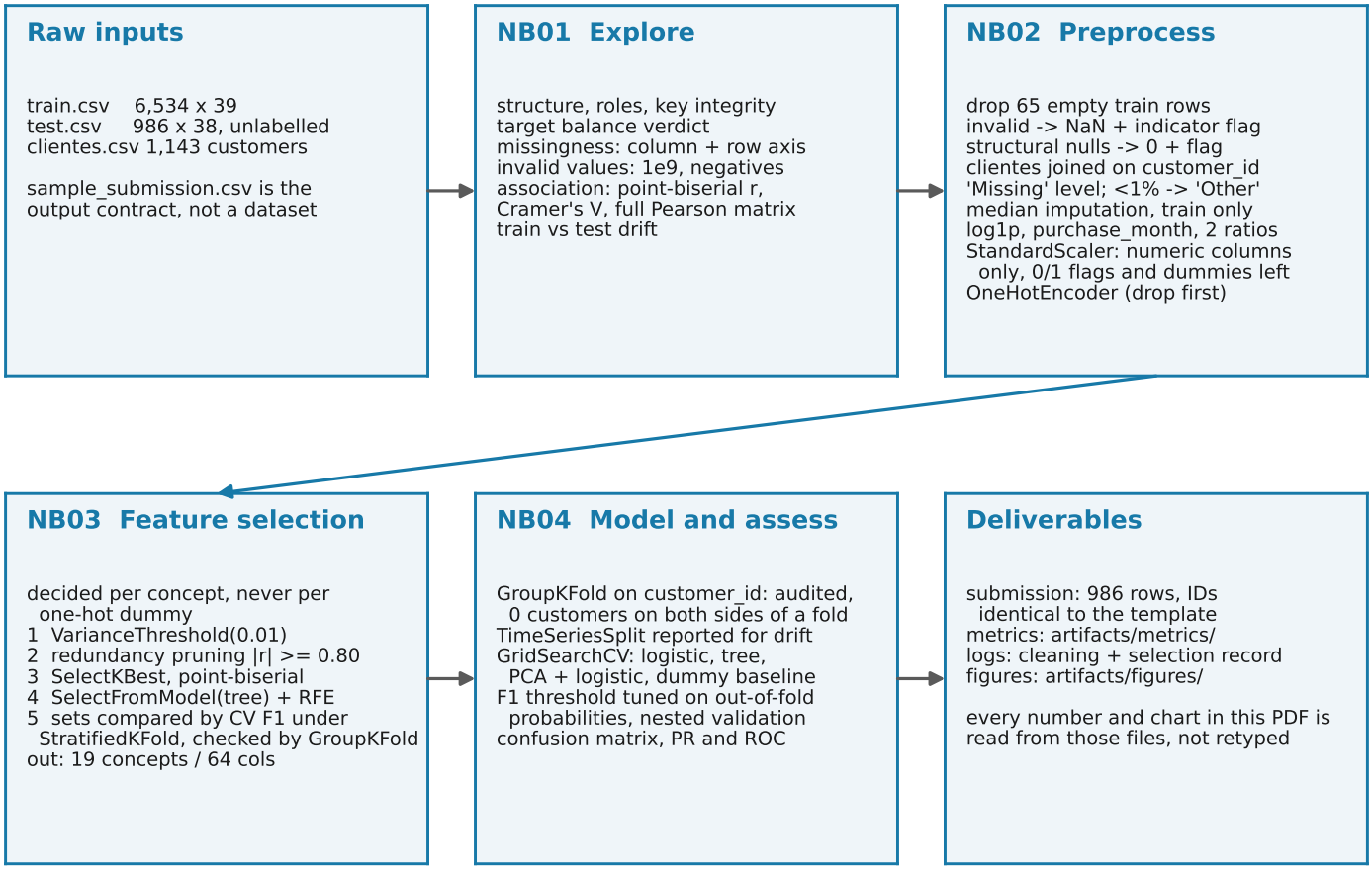
# Repeat Purchase Prediction - Pipeline Structure

Group 22 | Data Mining, NOVA IMS | Everything & Then Some | scored on F1 by Kaggle

One row is one purchase transaction: 6,534 training and 986 test rows over 1,143 customers, predicting repeat_purchase_90d. Classes are near balanced (56.0% positive), so the majority-class F1 of 0.718 is the floor.

## 1. Pipeline schematic




## 2. Why each step is appropriate

| Stage | Choice | Why it is appropriate here |
|---|---|---|
| Explore | course sequence: structure, then target, then quality, then association | roles and null patterns must be known before any statistic or transform can be trusted |
| Preprocess | invalid values become NaN plus a flag; structural nulls become 0 plus a history flag | a 1e9 placeholder is not a large purchase, and 1,048 rows belong to customers with no history - clipping or median-filling would invent data in both cases |
| Preprocess | standardisation is applied to the numeric columns only; 0/1 flags and one-hot dummies keep their scale | a dummy present on half a percent of the rows has a standard deviation of 0.07, so standardising it turns a single 1 into a value near 14 and lets that column escape an evenly applied penalty - measured in notebook 04 as a loss of ROC-AUC |
| Integration | clientes joined on customer_id; only attributes with <= 40 levels kept | the join is verified 1:1 so metadata cannot duplicate rows; commercial region reaches 38.5 points off the baseline, so it is kept and settled by ablation |
| Select | concept-level funnel; the final set is chosen by measured CV F1 under a stated parsimony rule | dummies of one category are not independent variables, and a rule stated in advance replaces a taste-driven cut |
| Model | logistic regression, GroupKFold, threshold tuned for F1 | customers repeat across rows, so grouped folds remove an optimistic split - the audit counts 0 customers and 0 rows on both sides of a GroupKFold fold, against 97% of validation rows having a known customer under a random split; F1 is the scored metric and is not maximised at 0.50 |
| Assess | nested threshold tuning plus a chronological holdout, split by customer type | the test window is later than train, so a random-CV number overstates what the leaderboard will see; and 94.5% of test rows belong to a customer already in train, which a grouped fold can never represent |

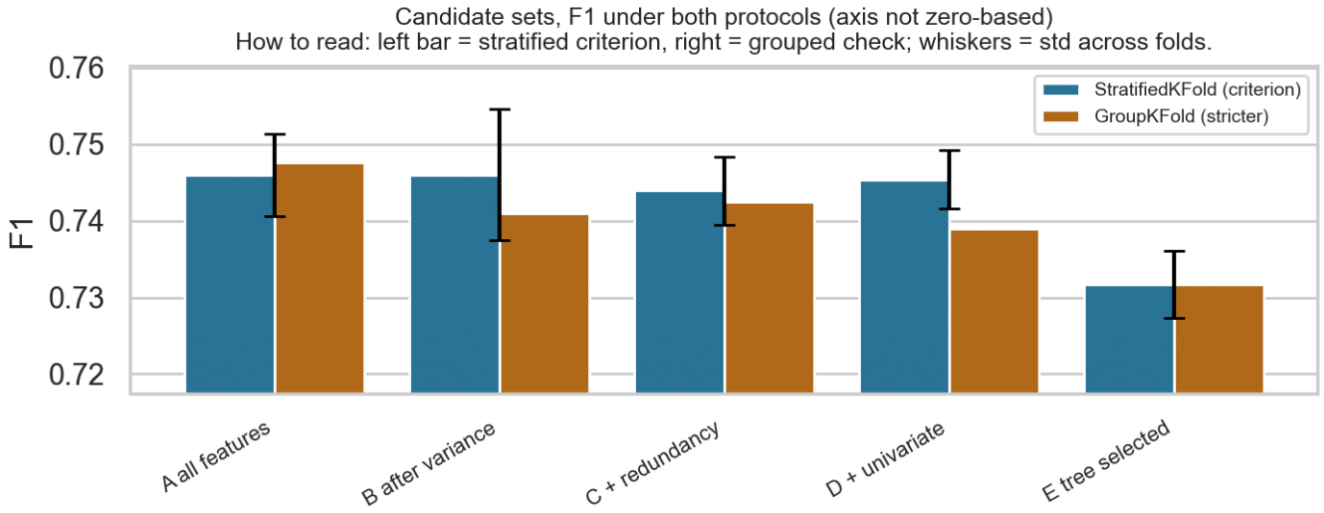
Reproducibility: install from requirements.txt or environment.yml, place the four supplied CSVs in data/, and run notebooks 01 to 04 in order from inside 01_notebooks/. The whole chain takes about a minute, writes only CSVs (no pickle), creates no folders, and never fits anything on test rows. RANDOM_STATE = 42 throughout, so every number on these two pages is reproducible from the code.

# Pipeline evidence - the notebooks' own figures

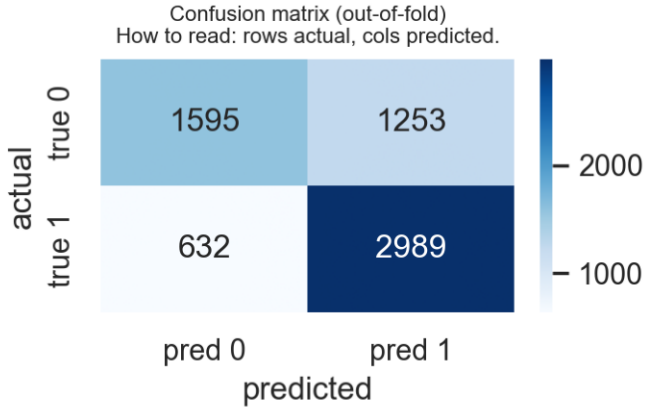
Group 22 | Data Mining, NOVA IMS | the charts below are the notebooks' own figures

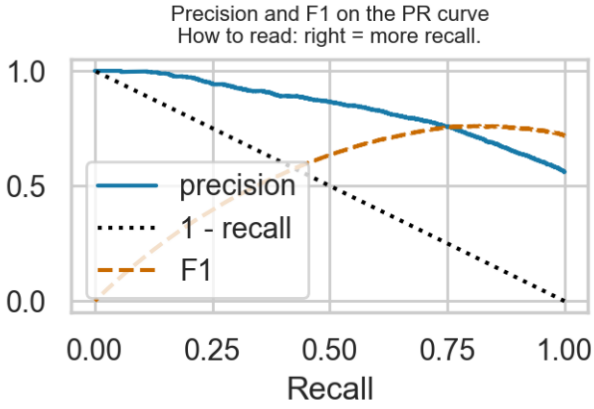
## 3. Feature selection: 53 concepts / 124 columns in, 19 / 64 out

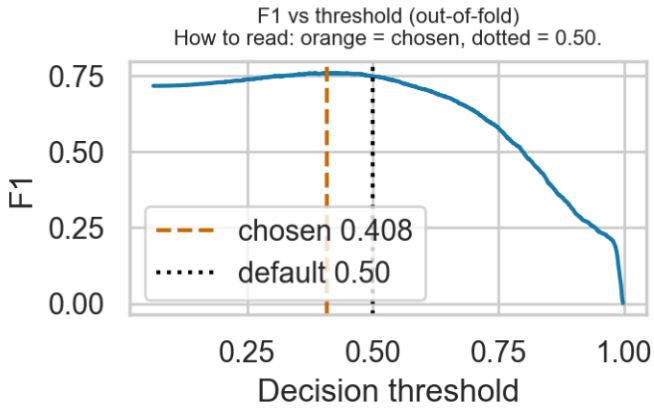
Ranked by measured F1 under StratifiedKFold, with GroupKFold on customer_id beside each set; the parsimony rule stated in advance settles the disagreement between them.

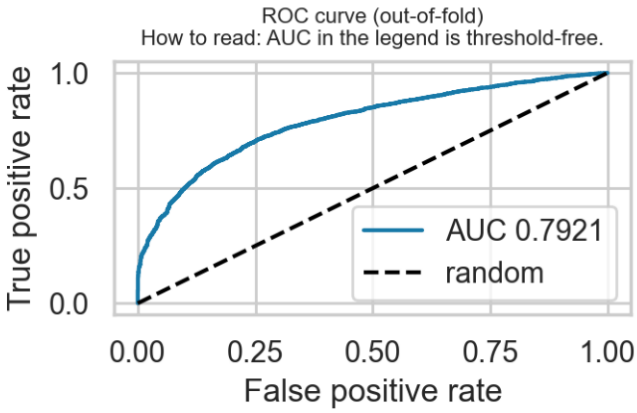



## 4. Model and assessment, out of fold




## 5. Headline numbers, submission and limitations

| Quantity | Value | Quantity | Value |
|---|---|---|---|
| Model | logistic, C: 0.03, class_weight: None | Grouped CV F1 | 0.7484 |
| Features | 124 columns | OOF F1 at 0.408 | 0.7603 |
| Nested honest F1 | 0.7550 +/- 0.0127 | ROC-AUC | 0.7921 |
| Chronological holdout | 0.7945 | Always-majority F1 | 0.7177 |
| Holdout returning / new | 0.8013 / 0.7296 | Submission positive rate | 76.5% |

Submission: 986 rows, ID and repeat_purchase_90d, IDs identical to the template and in order, values strictly 0/1, no nulls. Expected leaderboard F1 lies between the nested grouped 0.755 - every scored customer unknown - and the chronological holdout 0.795, the closer analogue since 94.5% of test rows belong to customers already in train.